A Experimental details

All experiments were performed using an NVIDIA A6000 GPU for accelerated computation. The models were implemented in Python and imported from Huggingface with PyTorch, and standard libraries like NumPy and Matplotlib were used for data processing and visualization. We evaluated the following models:

Model	Variants	Ref.
Pythia	2.8B	(Touvron et al., 2023)
LLaMA	2.7B, 3.1-8B, 3.2-1B	(Touvron et al., 2023)
GPT-2	Large-1.5B	(Radford et al., 2019)
Mistral	7B	(Jiang, 2024)

Table 4: Models evaluated in the experiments.

Whenever possible, results were reported as the average of three runs, along with the standard deviation (std). For experiments where repeated runs were not feasible, the random seed was fixed to 42 to ensure reproducibility.

B Additional experiments

B.1 Layer-wise PLS analysis

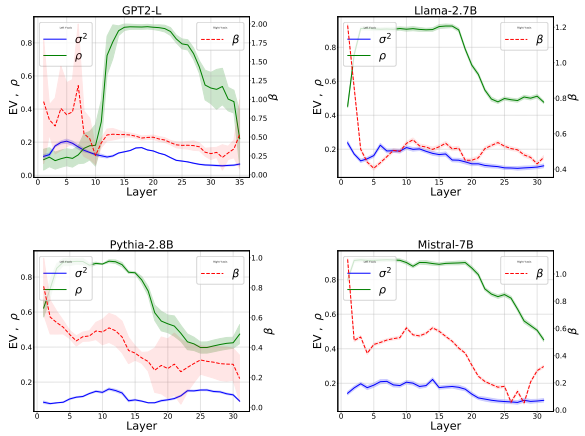

Figure 8: Layer-wise analysis of four models on letters groups, showing explained variance (σ^2), monotonicity (ρ), and Scaling Rate Index (β).

B.2 Birth year and population datasets projections in all layers

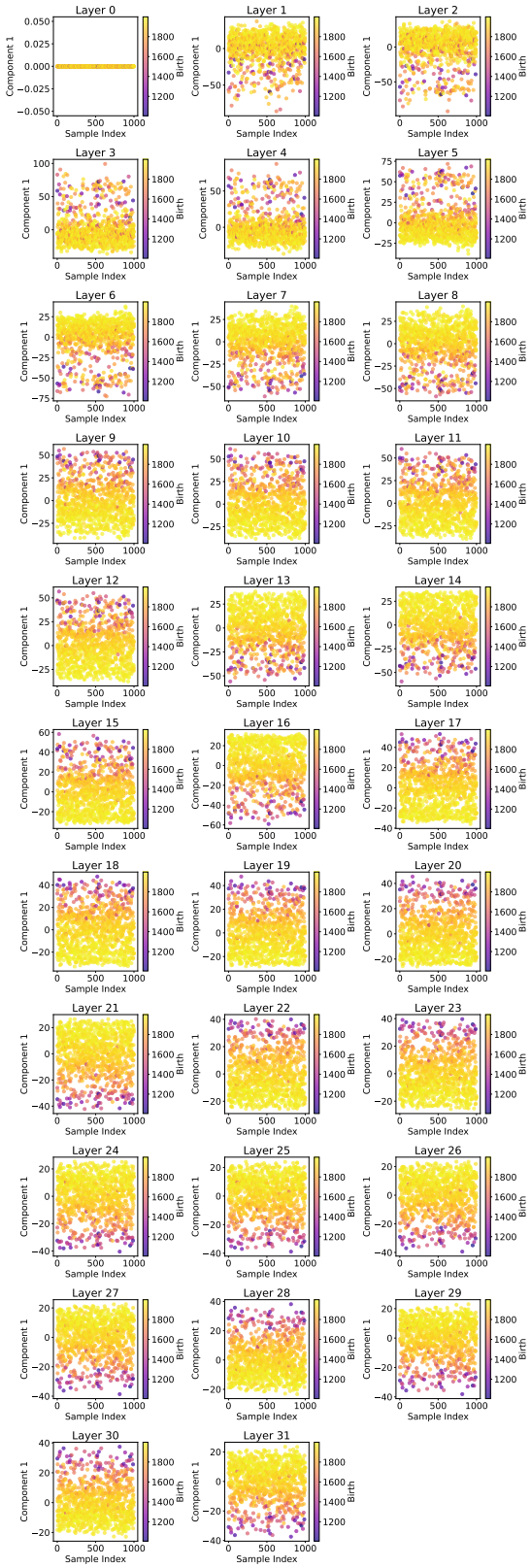

Figure 9: One component PLS model trained on Llama-3.1-8B instruct model activations to predict entities' birth year.

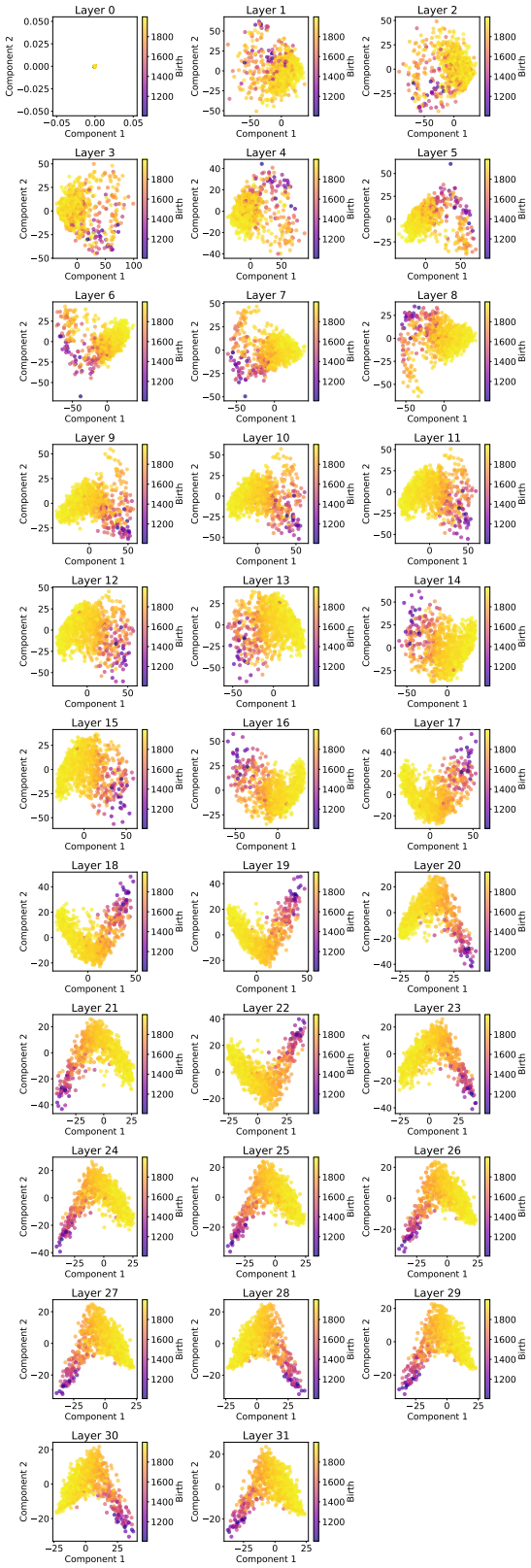

Figure 10: Two components PLS model trained on Llama-3.1-8B instruct model activations to predict entities' birth year.

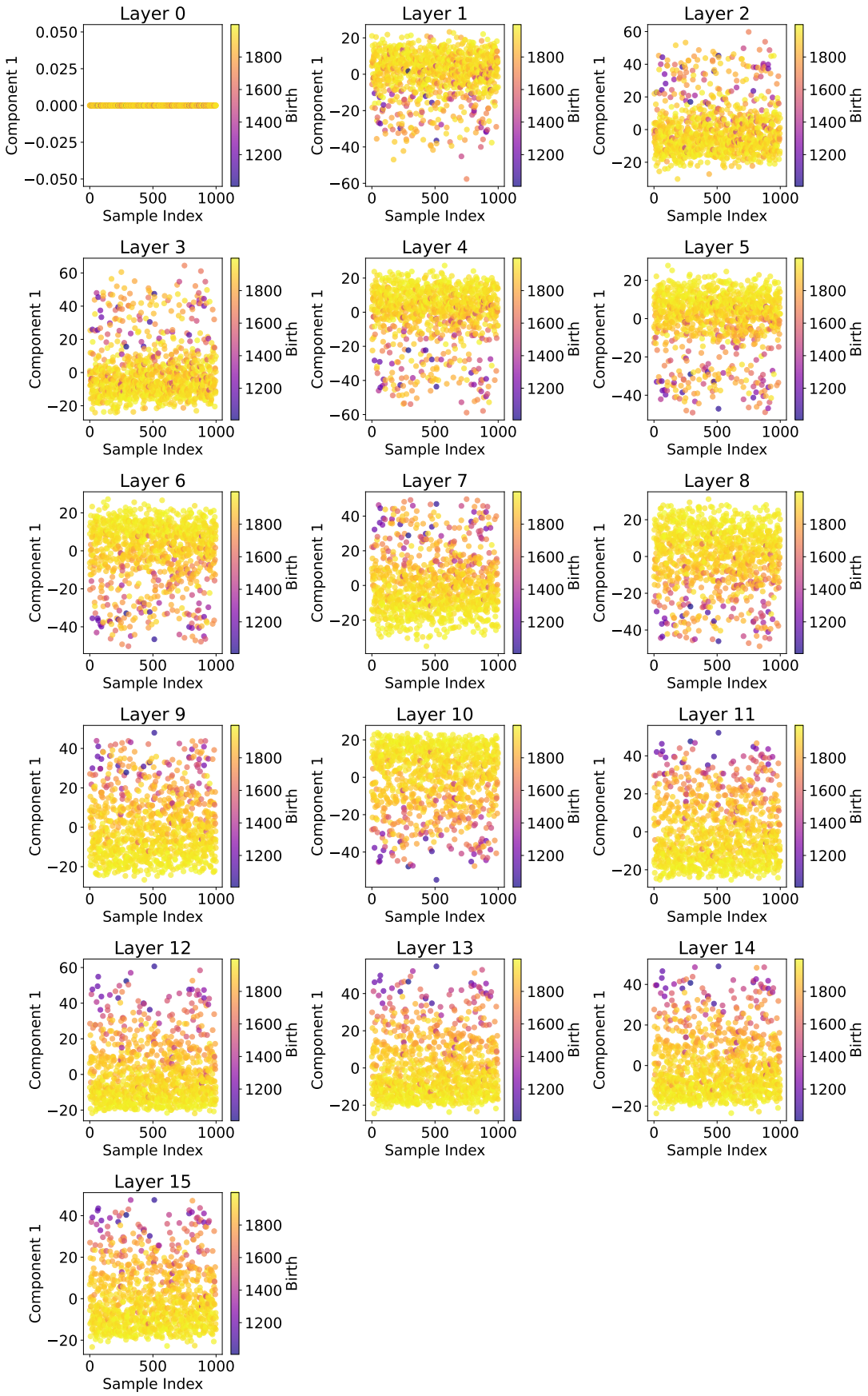

Figure 11: One component PLS model trained on Llama-3.2-1B instruct model activations to predict entities' birth year.